Material Handling concepts

March 2026

W Kruger

Introduction

FRED HILL
Material Handling Experts
Since 1894

800-523-0112
www.fred-hill.com

DHL logistics and their warehouses

Importance of Material Handling

- Material refers to commodities, information, people
- Material handling accounts for:
  - 25% of employees in a typical factory
  - 55% of factory space
  - 87% of production time
- An effective materials handling system can reduce operating costs by 15-30%
- Between 3-5% of material handled becomes damaged
- Causes a lot of industrial accidents
- Objective should be to eliminate & simplify material handling activities
- Important influence on operations design and layout design

Relationship between MH & facility layout

- The relationship between the two design elements involve:

  - Common data required for the design of each activity
  - A common goal to minimize costs
  - Both designs determine the effective use of space
  - Both determine the flow patterns in the facility

- Ideal approach - solve both problems **simultaneously**
- Complexity in practice – requires sequential design with iterations

Definitions of Material Handling

Definition 1

Material handling includes all the basic operations involved in the movement of bulk material, unit loads and individual products in any form, solid, liquid or gas, by means of machinery or by hand, within the boundaries of a place or business

Definition 2

Material handling is the art and science of moving, storing, protecting and controlling material

Definition 3

Materials handling means providing the right amount of the right material at the right place, time, sequence, condition, position and cost by using the right methods

Objectives of Material Handling

If designed well it can:

Increase throughput,

reduce inventories and operating costs

How to start this as part of your facility design?

1. Investigate current handling practices (use Audit sheet Table5.1)
2. Investigate and map process flow (distinguish **O**perations, **T**ransport, **S**torage and **I**nspection)
3. Complete Material Handling Planning chart (Fig 5.2) p185
4. Design total integrated 'system' (not just equipment)
5. Investigate equipment options last (keep track of new developments)

Auditing current Material Handling processes

Table 5.1 *Material Handling Audit Sheet*

		Condition Will Require:				
Conditions Indicating Opportunities for Improvement	Condition Observed	Supervisor Attention	Management Attention	Analytical Study	Capital Investment	Other Comments
1. Production equipment idle due to material shortage						
2. Material piled directly on floor						
3. In-plant containers not standardized						
4. Operators travel excessively for materials and supplies						
5. Excessive demurrage						
6. Misdirected material						
7. Backtracking of material						
8. Automatic data collection system not used						
9. Excessive trash removal						
10. System not capable of expansion and/or change						
11. No pre-kitting of work						
12. Crushed loads in block stacking						

Material Handling planning chart

MATERIAL HANDLING PLANNING CHART

Company A.R.C. Inc, | Prepared by I.A. | Layout Alternative 1

Product Air Speed Control Value | Date | Sheet 1 of

Step No.	O	T	S	I	Description	Oper No.	Dept.	Cont. Type	Size	Wt.	Qty. Per Cont.	Freq	Dist	Method of Handling
1			X		Bar stock in storage (2200)		Stores.							
2		X			Profit stores to saw dept.			LDDSE (FK.TRK)	2.5″ × 3.5 × 16″	5 lb	to bars	3 times daily	16 ft	Fork lift
3			X		Store in saw department		Saw							
4	X				Cut to length	0101	Saw							
5		X			From saw to grinding			TOTE pan	15″ × 12″ × 7″	30 lb	30	Twice daily	10 ft	Platform hand truck
6			X		Store in grinding		Grinding							
7	X				Grind to length	0201	Grinding							
8			X					TOTE pan	15″ × 12″ × 7″	30 lb	30	Twice daily	13 ft	Platform hand truck
9				X	Store in deburring		Deburring							
10	X				Deburr	0301	Deburring							
11		X			From deburring to dr. Prs			TOTE pan	15″ × 12″ × 7″	30 lb	30	Twice daily	16 ft	Platform hand truck
12			X		Store in drill press		Drill Press							
13	X				Dr. CD holes tap. rean, dsk	0401	Drill Press							
14		X			From dr. press to tur. lathe			TOTE pan	15″ × 12″ × 7″	30 lb	30	Twice daily	33 ft	Platform hand truck

Figure 5.2 Material handling planning chart for an air flow regulator. Key: Operation—O. Transportation—T. Storage—S. Inspection—I.

Engineering Design Process as applied to MH p181

1. <u>Define</u> the objectives and scope of the material handling system
2. <u>Analyze</u> the requirements for moving, storing, protecting and controlling material
3. <u>Generate alternative </u>designs for meeting material handling system requirements
4. <u>Evaluate</u> alternative material handling systems
5. <u>Select</u> the preferred design for moving, sorting, protecting and controlling material
6. Implement the preferred design including the selection of suppliers, training of personnel, installation, debug and start-up equipment and periodic audits of system performance

MHS Design Factors to be considered

- Capital requirements concerning equipment and unitizing of unit loads, availability of funds
- Physical properties of facility and available space
- Management policy in terms of safety and working environment
- **Interdependence** between handling and processing

MHS Design Equation

Material + Movement + Methods = Recommended Systems

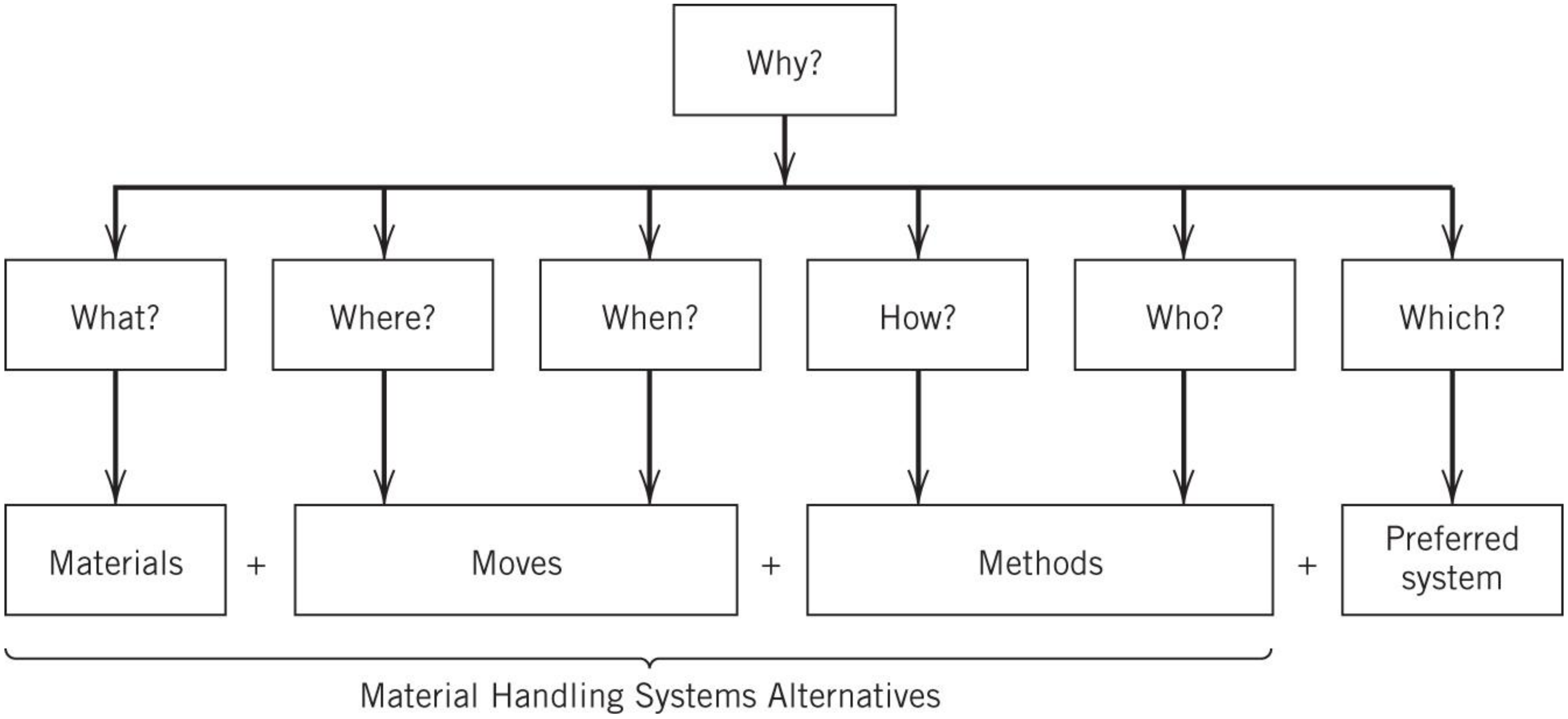


Figure 5.1

Problems In the Design of a MHS

- Interdependency between materials handling and layout
- **Wide variety of available equipment!!!! See p204 for classification and Appendix 5B for examples**
- Some characteristics of equipment are difficult to quantify (for example flexibility, mobility and effort)
- Objectives other than minimum cost (utilization, safety, maintenance, ease of expansion etc.)
- Some information required for the design of the system (e.g. movement time and equipment running cost) cannot be determined accurately
- Application of an analytical model usually requires complex calculations with no guarantee for an optimal system design e.g. p194-202

Characteristics of an efficient MHS

- Combines handling with processing where possible
- Automation where possible
- Safe
- Provides protection for material
- Standardization in types of equipment
- Maximum utilization of equipment
- Minimum backtracking, cross flows and handling
- Minimum congestion or delays
- Economical

20 Material Handling Principles

1. **Orientation**
   - Study the problem to identify existing methods, problems and physical and economical constraints to establish future requirements and goals
2. **Planning**
   - Establish a plan to include basic requirements, desirable options and the consideration of contingencies for all material handling storage facilities
3. **Systems**
   - Integrate economically viable handling and storage facilities into a coordinated system of operations
4. **Unit load**
   - Handle product in *as large unit loads as practical*/economical

Material Handling Principles (cont)

5. **Space utilization**
   - Make effective use of all *cubic* space
6. **Standardization**
   - Standardize methods and equipment where possible
7. **Ergonomic**
   - Recognize human capabilities and limitations by designing material handling equipment and procedures for an effective interaction with people using the system
8. **Energy**
   - Include energy consumption of material handling systems and material handling procedures when making comparisons or preparing economic justifications

Material Handling Principles (cont)

9. **Ecology**
   - Use equipment and procedures that minimize adverse effects on the environment
10. **Mechanization**
    - Mechanize where feasible to increase efficiency and economy
11. **Flexibility**
    - Use methods and equipment that can perform a variety of tasks under a variety of operating conditions
12. **Simplification**
    - Eliminate/reduce/combine movement and equipment
13. **Gravity**
    - Utilize gravity to move material where possible

Material Handling Principles (cont)

14. **Safety**
    - Provide safe equipment and methods (OHS act)
15. **Computerization**
    - Consider computerization in handling and storage systems for improved material and information flow
16. **System flow**
    - Integrate data flow with physical flow
17. **Layout**
    - Prepare an operations sequence and equipment layout for all viable system solutions – then select the alternative system which best integrates efficiency and effectiveness

Material Handling Principles (4)

18. Cost

- Compare economic justification in equipment and methods as measured by expenses per unit handled

19. Maintenance

- Prepare a plan for preventative maintenance and scheduled repairs on material handling equipment

20. Obsolescence

- Prepare a long-range and economically sound policy for replacement of obsolete equipment methods with consideration of after-tax life cycle costs

Mechanization of Material Handling scale

1. Handling by hand - dependent on physical strain

2. Mechanized handling replace human handling

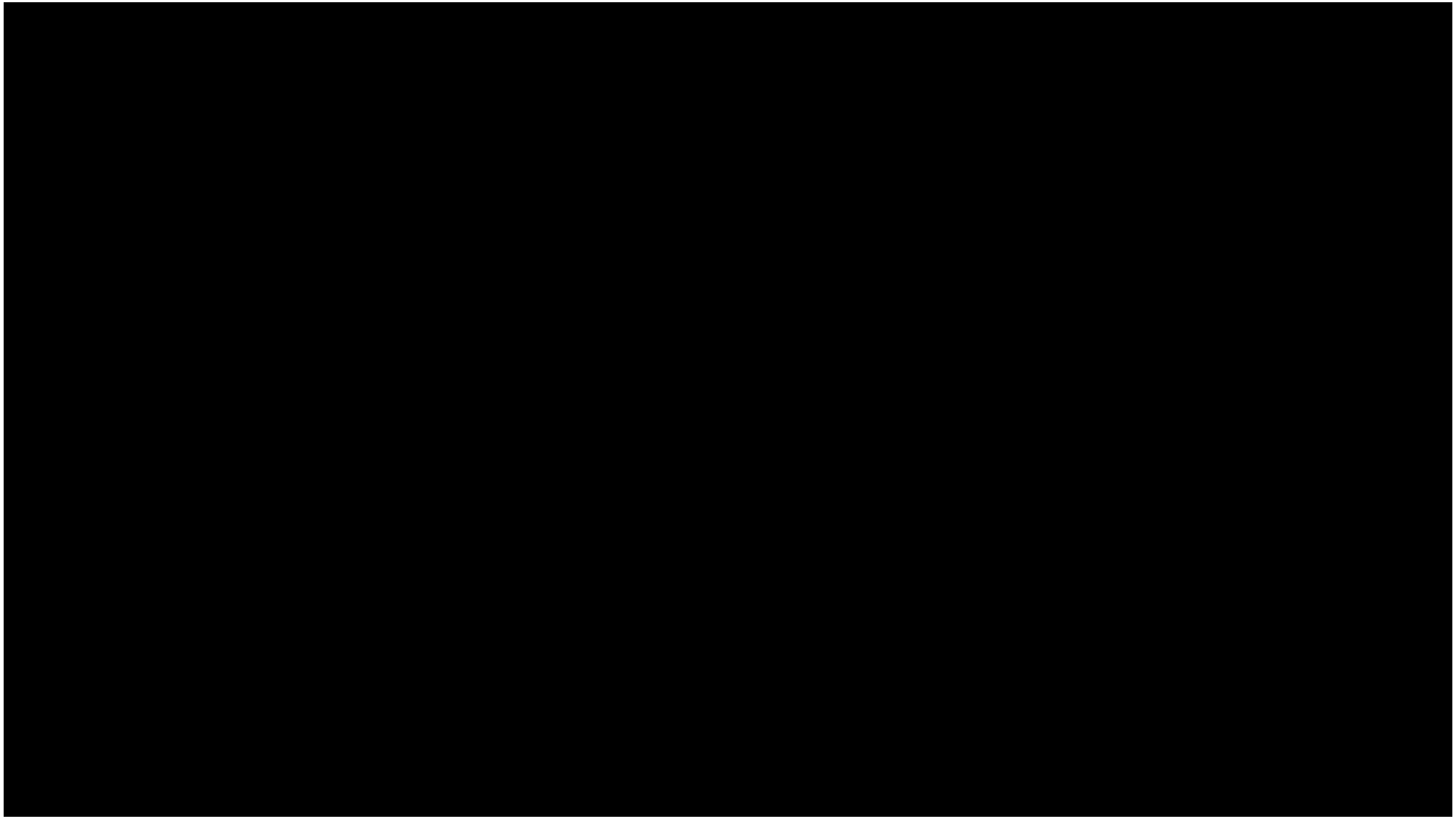
3. Mechanization complemented by computers - generate documents and specify movements and processes

4. Automated – minimal human intervention (BMW)

5. Fully automated – computer supply real-time control, no human intervention

DO YOU SEE THE IMPORTANCE OF THINKING PROCESS?

Fully Automated Factory BMW



Cost of Material Handling

- Initial cost of equipment and training
- Operating and Maintenance costs
- Purchasing price for unitizing material of unit loads
- Cost of packaging material
- Cost of material damages

Reduce costs how

- Minimize standing time
- Minimize back flow/double handling
- Reduce handling distances
- Plan maintenance procedures
- Correct equipment reduces damages
- Eliminate unsafe practices
- Standardize
- Use gravity where possible
- Replace obsolete equipment
- Utilize unit loads where possible

Unit Load design p187

- What is a UNIT? A single item, or items grouped together or bulk material that is arranged and restrained so it be *stored*, *picked up* and *moved* as a single mass

- The movement defines a unit load, and this specification determines material handling system design

- A unit load affects trade-offs – schedules, transport, WIP job completion

- Capability to handle more items at a time thus reducing the number of trips, handling costs, loading and unloading times and product damage

Factors Determining the Size of Unit Load

- Type, mass, size and shape of materials/items to be handled
- Quantity and frequency handled
- Susceptibility to damage
- Extent of handling (distance etc.)
- Environment (available space, aisle etc.)
- Reconcilability with the total logistics system
  - Other unit loads in system
  - Material handling and transport equipment in system
  - Method of receiving, storing, shipping and handling
- Cost of unitizing load
- Additional processes in unitizing a unit load such as stacking and protection materials

Factors Determining Equipment Choice

Material

- Type, size, volume, shape, weight, unit load of materials

Movement requirements

- Distance, frequency, path, aisle width, loading/unloading mechanism

Storage requirements

- Area, height, shape and obstructions in storage facilities, storage equipment and policies

Cost

- Capital and operating cost of equipment, interest rates, depreciation and lifespan

Other factors

- Flexibility and obsolescence

Equipment classification p204

1. Containers and unitizing equipment :
   - Containers – **<u>Pallets (see p192)</u>**, skids and skid boxes, tote pans
   - Unitizers – Straps/stretch-wrap, palletizers

2. Material transport equipment:
   - Conveyers
   - Industrial vehicles
   - Monorails, cranes and hoists

3. Storage and retrieval equipment:
   - Racks and stacks
   - Shelves, bins, cabinets

See both TEXTBOOKS for variety of equipment ideas

Tompkins App 5B
Stephens Ch 10 & 11